Industrial Internship Report on

”Project Name”

Prepared by

[Student name]

Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with the Industrial Partner, UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT regarding the development of an IoT-based solar system monitoring solution. We had to finish the project including the report in 6 weeks' time. The project involved creating a simulated monitoring environment as a software-based alternative to physical Data Acquisition and Monitoring Systems (DAMS). To achieve this, I utilized Node-RED for workflow orchestration and telemetry simulation, and Mosquitto as the MQTT broker to manage publish-subscribe communications for real-time data transmission. The final prototype enables live solar telemetry generation, real-time dashboard visualization of critical parameters (such as voltage, current, power, and temperature), threshold-based fault detection, and alert generation. This internship provided an excellent opportunity to gain practical exposure to industrial IoT challenges, communication protocols, and monitoring architecture, allowing me to design and implement a functional, simulation-based solution. It was an overall great experience that significantly enhanced my understanding of IoT system design and industrial data workflows.

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1 Preface

1.1.1 Internship Overview and Career Development

The industrial internship program, facilitated by upskill Campus and The IoT Academy in collaboration with the industrial partner, UniConverge Technologies Pvt Ltd (UCT), provided a critical bridge between academic theory and real-world engineering application. Engaging in this six-week program was essential for my career development, as it transformed conceptual knowledge of IoT architectures into practical, industry-standard skill sets. Understanding the necessity of such internships is paramount for students, as they provide exposure to professional workflows, problem-solving methodologies, and the technical rigors expected in the modern engineering landscape.

1.1.2 Project Definition and Scope

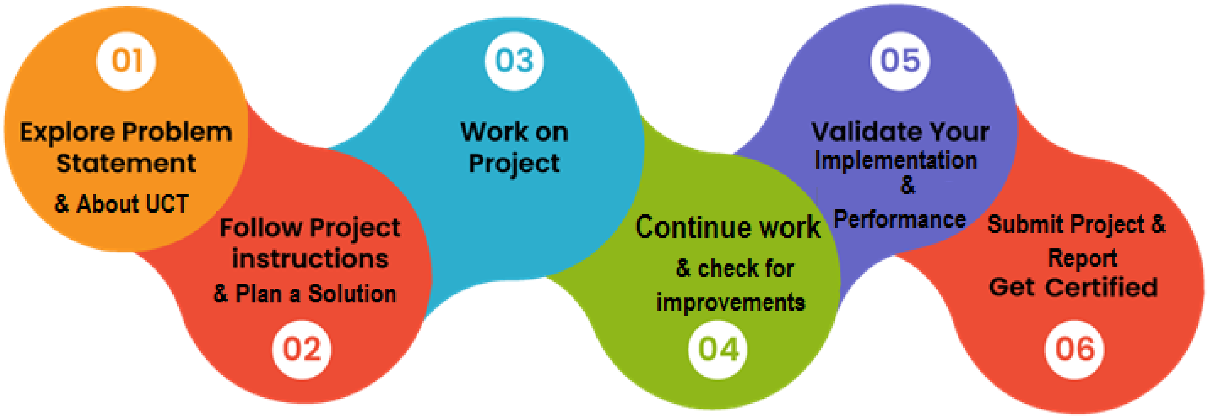
The problem statement provided by UCT focused on the development of an IoT-based Solar System Monitoring Solution. The primary challenge was to design a scalable monitoring framework capable of acquiring, processing, and visualizing data from a solar array. Because physical hardware access was limited, the scope was redefined to build a high-fidelity software simulation of a Data Acquisition and Monitoring System (DAMS), ensuring the architecture remained compatible with real-world sensor integration.

1.1.3 Program Structure and Execution

The program was meticulously planned as a six-week progression, moving from foundational protocol studies to complex system implementation.

- Weeks 1–2: Focused on understanding IoT ecosystems, MQTT communication protocols, and industrial dashboarding requirements.
- Weeks 3–4: Dedicated to the implementation of the simulation environment using Node-RED for workflow orchestration and Mosquitto as the MQTT broker.
- Weeks 5–6: Centered on testing, threshold-based fault detection, and finalizing the real-time visualization dashboard.

The six-week progression plan followed as given by USC/USC was:



1.1.4 Learnings and Personal Experience

This internship was a transformative experience that allowed me to gain hands-on expertise in telemetry simulation and real-time data visualization. By overcoming challenges related to data latency and message brokering, I developed a deeper appreciation for the reliability required in industrial IoT systems. I am incredibly grateful to the mentors at upskill Campus, the faculty at The IoT Academy, and the engineering team at UniConverge Technologies for their constant guidance and support throughout this journey.

1.1.5 Message to Peers and Juniors

To my fellow peers and juniors: embrace projects that challenge your understanding of systems architecture. Theoretical knowledge serves as the foundation, but technical proficiency is forged through the implementation of real-world problems. Always prioritize understanding the "why" behind communication protocols and data flows; this depth of knowledge is what distinguishes a proficient engineer.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

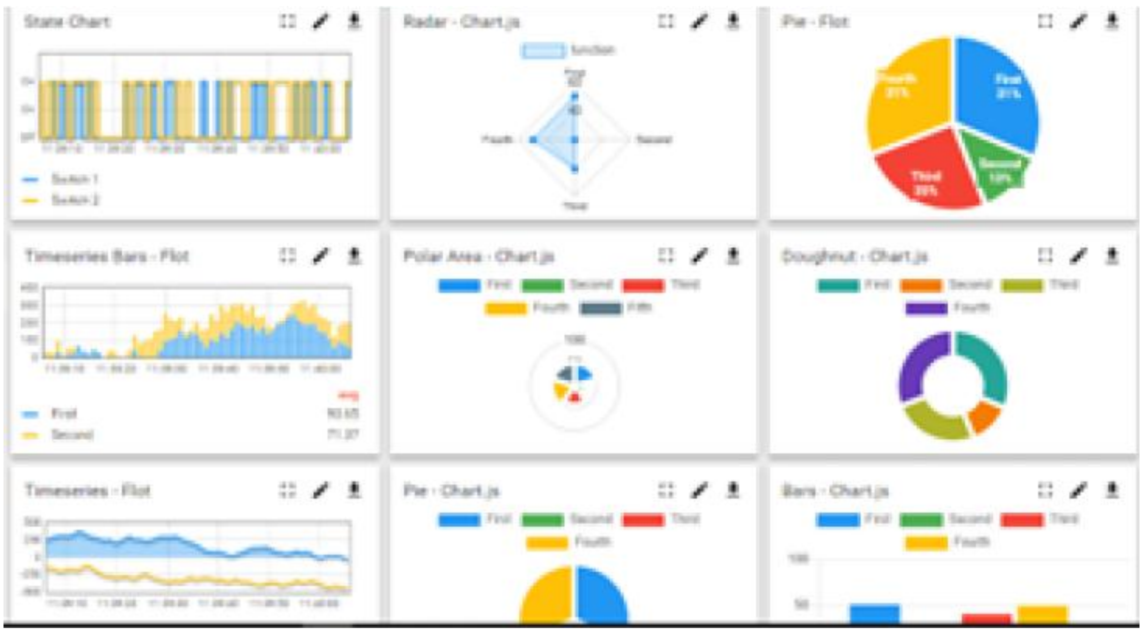
i. UCT IoT Platform (uct Insight)

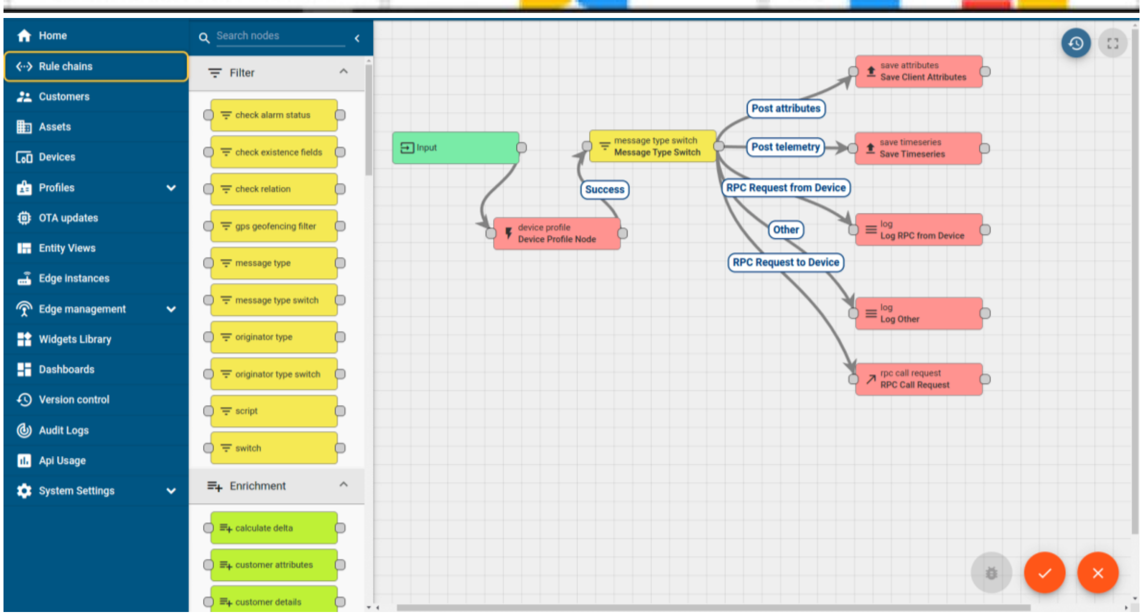
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
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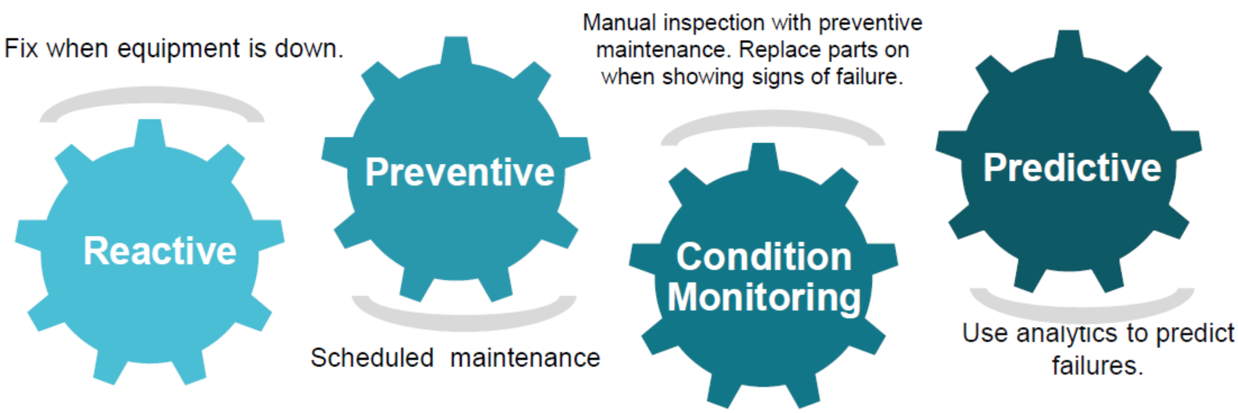
iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

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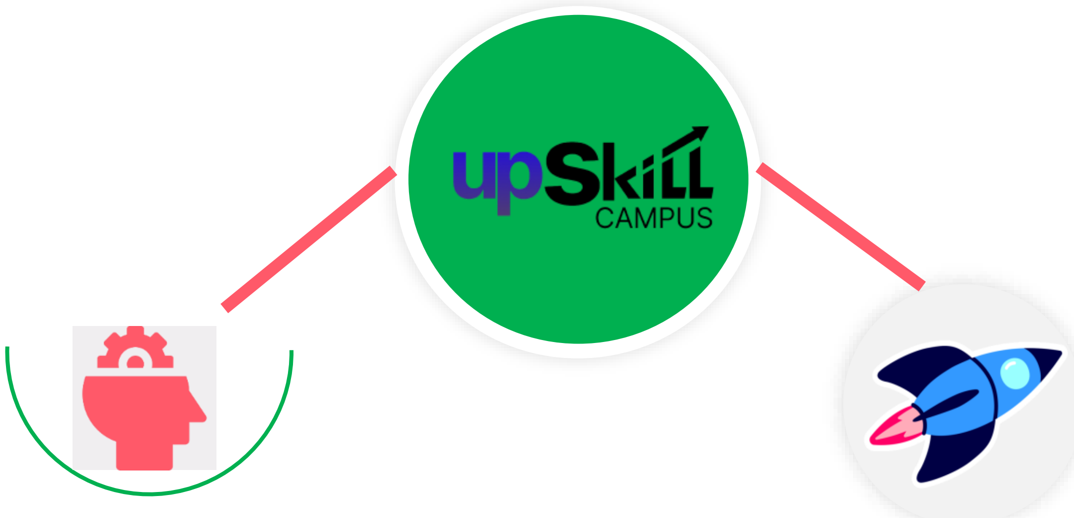
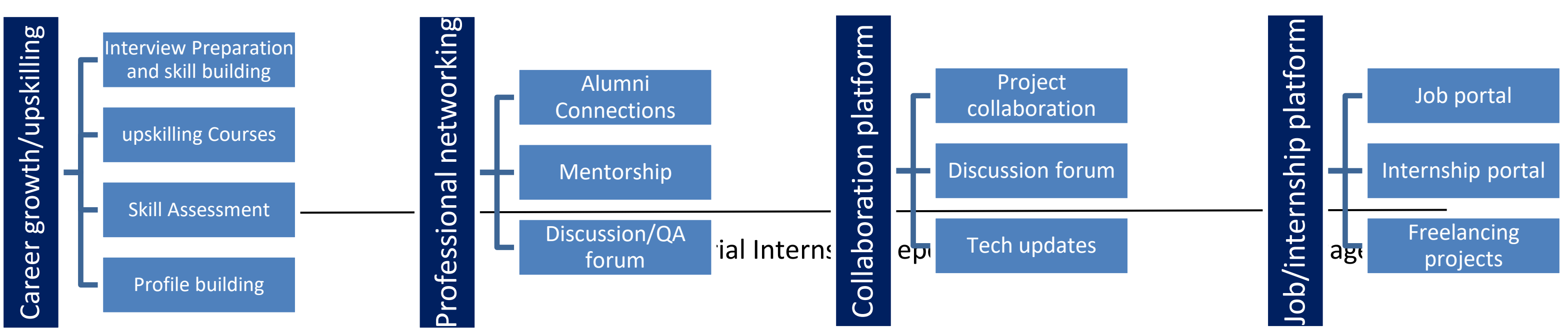


Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1]

[2]

[3]

2.6 Glossary

Terms	Acronym

3 Problem Statement

In the assigned problem statement

[Explain your problem statement]

4 Existing and Proposed solution

Provide summary of existing solutions provided by others, what are their limitations?

What is your proposed solution?

What value addition are you planning?

4.1 Code submission (Github link): CODE SUBMISSION LINK

4.2 Report submission (Github link) : first make placeholder, copy the link.

5 Proposed Design/ Model

Given more details about design flow of your solution. This is applicable for all domains. DS/ML Students can cover it after they have their algorithm implementation. There is always a start, intermediate stages and then final outcome.

5.1 High Level Diagram (if applicable)

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)

5.3 Interfaces (if applicable)

Update with Block Diagrams, Data flow, protocols, FLOW Charts, State Machines, Memory Buffer Management.

6 Performance Test

This is very important part and defines why this work is meant of Real industries, instead of being just academic project.

Here we need to first find the constraints.

How those constraints were taken care in your design?

What were test results around those constraints?

Constraints can be e.g. memory, MIPS (speed, operations per second), accuracy, durability, power consumption etc.

In case you could not test them, but still you should mention how identified constraints can impact your design, and what are recommendations to handle them.

6.1 Test Plan/ Test Cases

6.2 Test Procedure

6.3 Performance Outcome

7 My learnings

You should provide summary of your overall learning and how it would help you in your career growth.

8 Future work scope

You can put some ideas that you could not work due to time limitation but can be taken in future.